

LoRaWAN has the advantage of being an open platform to facilitate various enterprises entering the IoT application market. "Earlier we had Wi-Fi, Bluetooth and such systems, but we could not connect everything non-computer to the system. Here, we have a very simple sensor that can be attached and then we can connect the hardware to the system," added Chauhan. "Once you get through the initial phase of deployment, it gets easier to work with the system."

Interesting implementations

Some sensors, some connectivity and with proper embedded development skills, you could have your own IoT system. Applications include interfaces, connected cars, entertainment, residences and smart homes, wearable technology, quantified self, connected health and smart retail. As for LoRaWAN, areas of implementation focus on connectivity. IBM long-range signaling and control, or implementation by Tata Communications are just the tip of the iceberg. With proper implementation, the possibilities are endless.

With increased usage, availability of software is no more a concern. Software available on the Internet makes life easy in handling LoRa radios. Using HOPE LoRa

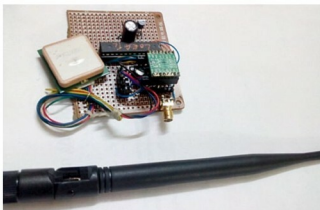


Fig. 2: LoRa transmitter used in the chlorination plant project

transceivers connected on serial peripheral interface bus, Somnath Bera, additional general manager, NTPC Ltd, solves a problem of indication in chemical plants. Command files in the software, for example, make working very easy on LoRaWAN. Bera says, "In less than ten lines one can make these radios up and running using available headers."

He explains the problem, "The status of chlorination plant chlorine leakage along with chlorine cylinder temperature was not available at the control room of the new chemical plant, which is situated 3.5km away from the chlorination plant." Since the control room of the new chemical plant is situated at some distance, reaching there would take some time. Using Wi-Fi at this range becomes meaningless. Requirement of bi-directional transfer of data between the control system and the chlorination plant

lead to the involvement of LoRaWAN.

Nationwide LoRa networks

Earlier this year, Netherlands became the first country to implement LoRaWAN, with the project taking about eight months. "Last year we identified an increasing demand for low-power network technology for IoT applications. We are responding to

this by choosing LoRa, so millions of devices can be connected to the Internet in a cost-effective manner," said Joost Farwerck, chief operations officer and member of the board of management, KPN, in a press release.

SK Telecom has also announced the availability of a nationwide LoRa network in South Korea this year. The firm offers 100MB of data per month for US\$ 1.75.

Another South Korean fixed-line telecom firm, KT, has plans to invest in NB-IoT network and offer free of charge 100,000 modules to developers. Might just be for the first year, it can certainly drive adoption.

There are also proof-of-concept projects underway, which could implement LoRaWAN in the future. Baggage handling, facility services and systems to monitor rail switches are some real-life examples.

Security is just one issue

Ryan D'Costa, business development manager, inventrom, says, "You do not want your data to fall into the wrong hands." Shukla says, "Since it is an open network, data security is a big question."

Shukla adds, "It might look lucrative with respect to cost and power, but industries cannot take the risk of data hacks, so currently they use LTE based gateways and industrial-IoT (IIoT) solutions."

Reports have highlighted some

What LoRaWAN is

The IoT deals with connectivity and sensors for the most part. In order to get this connectivity, several networks are being tested. Due to limitations, the common wireless technologies cannot be used efficiently. However, LoRaWAN is a system that allows range requirements higher than the traditional Wi-Fi system. With LoRaWAN, the range extends to kilometres. So a good solution to cover a city, maybe?

According to Link Labs, "LoRaWAN is a media access control layer protocol designed for large-scale public networks with a single operator." LoRa alliance, however, intends it for a much wider range of use. As per the company, "LoRaWAN is a low power wide area network specification intended for wireless battery-operated things in regional, national or global networks." LoRaWAN targets key requirements of the IoT, such as secure bi-directional communication, mobility and localisation services.